

The Statistical Significance of Statistical Insignificance. A Reproduction Study of ‘The Null Result Penalty’

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Abstract

Chopra et al. (2023) assesses whether scientific studies with null result findings are perceived as less publishable. Using a randomized controlled trial, the authors randomize a series of study vignettes to present statistically significant or statistically insignificant results. They present these vignettes to a sample of economists to judge the publishability and merit of these studies. They confirm that economists view null results as less publishable, as well as less important and having lower quality. We computationally reproduce their results using two statistical softwares – Stata and R. Despite being reproducible, we find a strange distribution of the primary outcome measure in the data which may explain the highly robust findings. To validate this study, we advocate for a replication of this study using a new sample of respondents and a separate set of researchers.

KEYWORDS: Economic Methodology, Communication, Peer Review

JEL CODES: B41, D83

1 Introduction

Studies that present null results are often perceived as providing less valuable evidence to the scientific community. While this conception is considered common, scarce evidence exists if these supposed widely held beliefs are acted upon in the peer review process. Chopra et al. (2023) tackle if such opinions influence the peer review process. The authors devise a randomized controlled trial soliciting opinions on the publishability of five study vignettes from 480 economists through an online survey. Using a factorial design, the authors randomize attributes about the vignettes which may influence the opinion of a reviewer on the likelihood that the study might publish. Critically, the authors design the vignettes to present results that show point estimates and standard errors (or p-values) which are statistically significant. But for half of the vignette-respondent pairings, the authors randomly shift the point estimates toward zero to create a null effect. In this way, Chopra et al. (2023) causally evaluate if presenting null results reduces the likelihood that a peer reviewer will deem that study publishable (e.g., a valuable contribution). The authors find that null results reduce the respondent’s belief that the study vignettes are publishable¹, as well as reducing their belief in the quality and importance of the study.²

We computationally reproduce these results using the original statistical software, Stata, as well as an alternative software, R. We find that the estimates are not qualitatively different across software. Exploring the distribution of the primary outcome measure – likelihood to publish – we note two oddities with Chopra et al. (2023). First, despite a small sample³, the reported results have an unusually large concentration of z-statistics distributed larger than 2.33 (e.g., statically significant at a 1 percent level). The distribution of the reported z-statistics are anomalous compared to other randomized controlled trials in economics. Second, we show that the distribution of likelihood of publication for the control group appears to be the mirror of the distribution for the treatment group (mirrored about a 50% likelihood of publication). This is an odd statistical artifact responsible for the robust statistically significant result in Chopra et al. (2023).

Because the distribution of the primary outcome measure does not present what would

¹“Table 3 shows the effects of the null result treatment on our main outcomes of interest. Panel A shows estimates controlling for respondent fixed effects, while panel B shows estimates excluding respondent fixed effects. As shown in column (1), respondents assigned to the null result treatment indicate that the studies have a 14.1 percentage point lower probability of being published (95% CI [−16.2, −11.9]; $p < 0.001$). This effect size corresponds to a 24.9% reduction in perceived publication chances.” (Chopra et al., 2023)

²“Respondents in the null result treatment associate the studies with 37.3% of an SD lower quality (95% CI [−49.6, −25]; $p < 0.001$), consistent with a mechanism in which researchers broadly associate studies that yield null results with lower quality... Respondents in the null result treatment associate the studies with 32.5% of an SD lower importance (95% CI [−43, −21.9]; $p < 0.001$) and think other researchers would associate the studies with 41.7% of an SD lower importance (95% CI [−52.6, −30.7]; $p < 0.001$).” (Chopra et al., 2023)

³“The 480 respondents in our main study complete four out of five vignettes, providing us with 1,920 observations in total. We, therefore, obtain 40 observations per factorial combination (median 40, min. 27, max. 54).” (Chopra et al., 2023)

be expected for an independent and identically distributed random sample, we propose a replication of Chopra et al. (2023) be conducted using new respondents and a separate set of researchers to validate these results. Our call for a replication study falls within the continued effort in economics to reproduce and replicate important scientific findings to improve the credibility of new findings and renew the public’s trust in the economic sciences (Brodeur et al., 2026a,b; McWay and Braaksma, 2025; McWay, 2025).

2 Narrow Replication: Computational Reproduction

Using the public replication packet⁴ for Chopra et al. (2023), we are able to computationally reproduce all results presented in this Economic Journal article as well as the supplementary appendix. Following a similar protocol as McWay et al. (2024); Kouchekinia et al. (2026), each member independently reproduced all tables and figures produced in the manuscript. Critically, we reproduce the main result claims presented in Table 3 of Chopra et al. (2023) using the original coding software Stata⁵ as well as in the R programming language⁶. The reproduction of Column 1 from Table 3 of Chopra et al. (2023) using Stata and R are presented in Columns 1 and 2 of Table 1, respectively. These estimates are qualitatively identical to the published results.

This study is computationally reproducible from raw data (CRR) as described by the Guide for Advancing Computational Reproducibility in the Social Sciences (ACRe) established by the Berkeley Initiative for Transparency in the Social Sciences (BITSS) (Bogdanoski et al., 2020). We assign the computational reproducibility of Chopra et al. (2023) as an Adjusted Level 10*⁷ within the BITSS 10 point scale. Table 2 provides an overview of the computational reproducibility of the manuscript as suggested by Rose et al. (2025).

3 Broad Replication: Re-analysis of Vignette Sub-samples

As a form of broad replication, we test the robustness of the results by re-analyzing Column 1 of Table 3 from Chopra et al. (2023) sub-setting the sample by the specific vignettes that respondents judged as publishable. We compare our estimates from these sub-samples in Table 1 for Columns 3 through 7 to the reproduced results pooling the responses across all five vignettes. The magnitude of the null treatment on publishability as well as the statistical significance of the estimated effects remain quantitatively similar

⁴The replication packet can be found on Zenodo: <https://zenodo.org/records/8168773>.

⁵We use Stata 16 SE to estimate results.

⁶We use R version 4.4.3.

⁷“Adjusted Level 10 (L10*): All requirements for Level 9* are met, and a certification is provided that the display item can be reproduced from the raw data (CRR). Level 10 (L10): All necessary materials are available and produce consistent results with those presented in the paper. The reproduction involves minimal effort and can be conducted starting from the analytic data (CRA) and the raw data (CRR).” (Bogdanoski et al., 2020)

across all vignettes. This provides assurance that the main results are not driven by statistical leverage from one vignette study particularly polarizing opinions across treatment status. Though it remains surprising that none of the vignettes exhibited an insignificant, or even modestly significant, effect simply by chance.

We present our re-analysis within the broader context of Chopra et al. (2023) through a specification curve (Simonsohn et al., 2020). Figure 1 demonstrates the variation in estimated coefficients and statistical significance between the preferred specification across all alternative specifications reported in the study as well as our re-analysis. This visualizes the variation in results considering the combination of analytical choices that produces each result. The preferred specification of the authors – indicated in red – is in the middle of the distribution of reported estimates. Of the 64 reported estimates and the 7 replicated estimates, only one estimate is marginally insignificant at a 5 percent level considering software used, the sample selected, fixed effects applied, vignette features considered, control variables, re-weighting schemes, and adjustment for multiple hypothesis testing.⁸ This figure demonstrates the resounding immutability of the findings. The lack of variation in the results is a bit extraordinary supposing that the results would eventually vary simply by chance. The specification curve motivates our exploration of the underlying outcome distribution across treatment status underpinning these results.

4 Z-Scores and the Distribution of Publishability

We note that Chopra et al. (2023) presents very robust results to several sensitivity checks. This is strange given that the sample size is relatively small to consistently estimate statistically significant result across several analytical choices. The authors note on page 203 that their study is sufficiently powered, “Using a clustered bootstrap procedure that resamples respondents with replacement, we estimate a minimum detectable effect size of 15.6% of an SD (corresponding to a 4 percentage point difference in perceived publication chances) at 80% power in our main specification and a significance threshold of 5%.” The estimated effect for the primary outcome of interest – publishability – presented in Table 3 Column 1 using the author’s preferred specification is -14.058 points on a 100 point scale. This is larger than the minimum detectable effect of a 4 percentage point difference estimated through their power calculation. Yet, the power calculation is designed to estimate the minimum size of the first moment in the distribution. Statistically, we should still expect that such a small sample should not *always* estimate statistically significant point estimates above this minimum detectable threshold given a normal distribution of possible treatment effects.

Focusing on the statistical significance of publishability, we compare the z-scores for

⁸The only insignificant result is the sub-sample analysis of respondents identifying from the field of economic theory in Figure 3 of Chopra et al. (2023).

the presented results in Chopra et al. (2023) against the distribution of z-scores in the economic literature provided by Brodeur et al. (2020). Figure 2 presents the distribution for both of these sets of z-scores. For most RCTs in economics, z-statistics are statistically insignificant (viz., less than a z-score of 2). By contrast, the z-statistics for publishability in both the manuscript as well as the supplementary appendix of Chopra et al. (2023) center around 8. It is ironic that a study testing the impact of a statistically insignificant null result on publishability presents such a resounding statistically significant finding. On average the z-statistics are four times larger than the average z-statistics for RCTs in economics. This contrasts the expectation that on average the authors should estimate statistically insignificant results – albeit with qualitatively similar estimated coefficients. Given the extensive sensitivity checks provided by Chopra et al. (2023), we do not believe there is selection about the published results. Rather, it appears that the composition of the sample used in this study is perfectly positioned to estimate statistically significant results despite the small effective sample of 480 respondent clusters resulting in 1,980 observations.

To further elaborate on the unique nature of the sample composition, we examine the distribution of perceived publishability. The outcome measure is a rating by respondents of how likely a presented vignette study is to publish along a sliding scale from 0 to 100. Figure 3 presents a histogram of the perceived likelihood of publishing each vignette distinguishing between if the respondent received a null finding (i.e., Low = 1) or a statistically significant finding (i.e., Low = 0). The reported effect of the null treatment is visually apparent when comparing the distribution of the treatment group (blue) to the control group (red). These distributions are nearly mirrors of one another about the 50:50 threshold for publishability. In the second sub-figure, we test this visually by overlaying the reciprocal⁹ for the control group’s responses onto the treatment group. Upon the transformation, we note that the distribution of publishability scores are almost identical for both treatment and control status. It is odd that the null treatment effect would have a reciprocal effect on publishability scores about the threshold 50:50. To test if these two distributions are in fact the same distribution, we perform a Kolmogorov-Smirnov test. This is a nonparametric test of the equality between one-dimensional probability distributions. With a test statistic of 0.0551 and a p-value of 0.1093 we cannot reject the null hypothesis that the distribution of publishability for the treatment group and the distribution of publishability for the reciprocal of the control group are the same. Rather, the outcome measure appear to be the same distribution flipped about the 50:50 threshold depending on treatment status. This is significant as it is the identifying variation that results in the estimated effect.

⁹For the control group, we apply the transformation: $publish = 100 - publish$

5 Conclusion

We computationally reproduce the results of Chopra et al. (2023) using two statistical softwares: Stata and R. While examining the distribution of the z-scores and responses by participants judging publishability, we find that the outcome measure has strange statistical properties across treatment status. This may explain why the statistical significance of Chopra et al. (2023) is distinctly more pronounced than would be expected given the distribution of z-scores for randomized controlled trials in economics. In the spirit of Mertonian Norms (Merten, 1942), we advocate for a replication of this study using a new sample of respondents and a separate set of researchers to independently verify these results. Given the minimal logistical hurdles to administer the study design and public access to the computationally reproducible replication packet, we believe that a replication of this study is not only warranted but feasible to attain. A replication of this study may confirm potential issues with the sample used in Chopra et al. (2023) or provide supporting evidence for the unique nature the population of researchers' opinions in the peer review process.

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6 Exhibits

Table 1: Computational Reproduction and Re-analysis by Sub-sample

	<i>Publishability</i>						
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Null result treatment	-14.058*** (1.090)	-13.988*** (1.097)	-15.063*** (2.476)	-18.105*** (2.611)	-14.921*** (2.487)	-15.154*** (2.578)	-10.437*** (2.578)
Mean Dep. Var.	57.128	57.128	57.124	58.059	53.788	64.022	52.672
Sample	Stata	R	Vignette 1	Vignette 2	Vignette 3	Vignette 4	Vignette 5
Observations	1,920	1,920	387	377	385	389	382
Adjusted R ²	0.5143	-0.070	0.199	0.224	0.132	0.129	0.080

Notes: This table reproduces Panel A estimates from Table 3 in Chopra et al. (2023). The first column is the original reported results in Column 1 of Table 3 (the main outcome of interest) computationally reproduced in Stata. Column 2 is the computational reproduction using R. Columns 3 - 7 re-analyze the result sub-sampling for each vignette. Standard errors are heteroskedastic robust clustered at the individual level. Statistical significance is represented by: *p<0.1; **p<0.05; ***p<0.01

Table 2: Computational Reproducibility

	Fully	Partially	No
Raw data provided	x		
Cleaning code provided	x		
Analysis data provided	x		
Analysis code provided	x		
Reproducible from raw data	x		
Reproducible from analysis data	x		

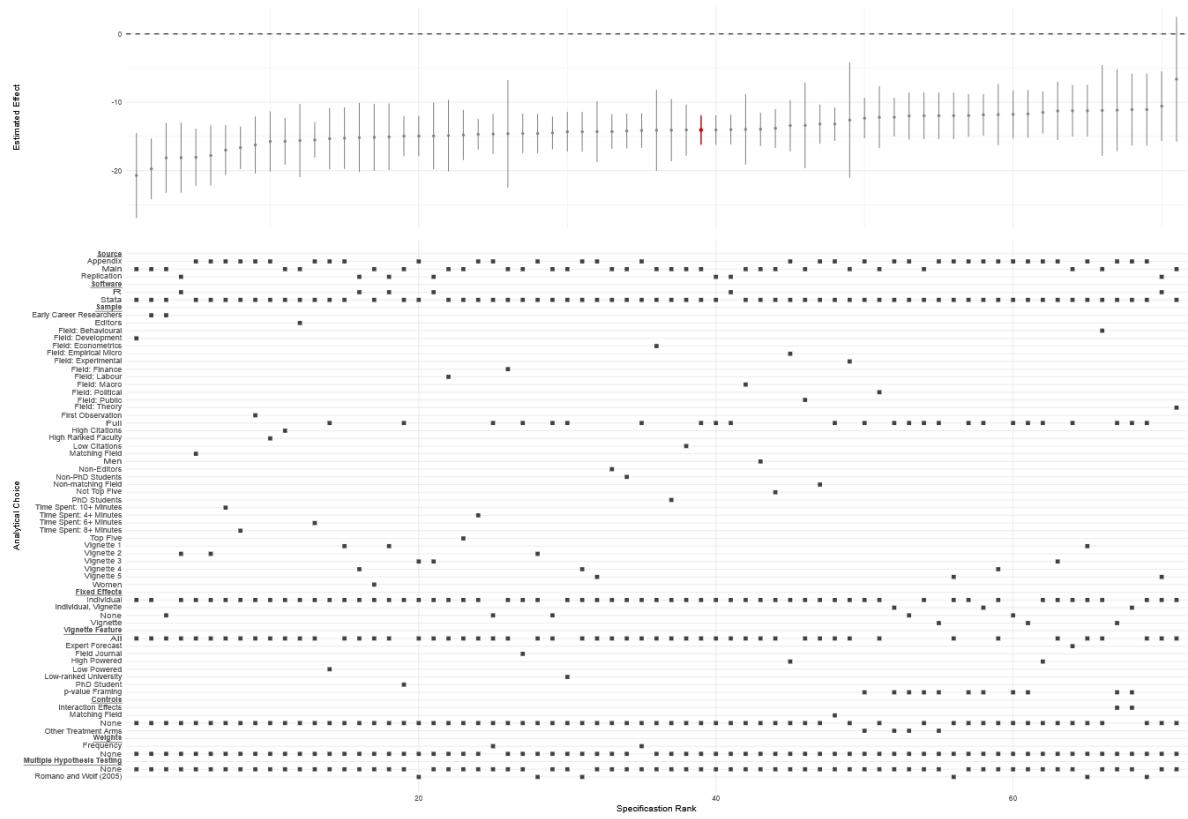
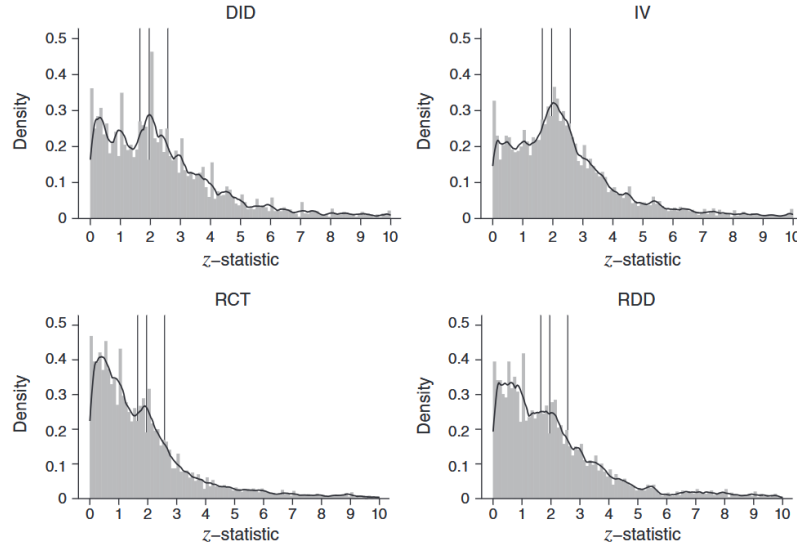


Figure 1: Specification Curve of Chopra et al. (2023) Results

Notes: This figure presents a specification curve where each column is a regression specification with the point estimate and the 95% confidence interval. The panel below the figure displays the combination of analytical choices that produces each result. The estimate in red is the preferred specification of the main result reported by the authors. In bold are the sections of the analytical decisions and the rows are the analytical selections.

(a) Distribution of Z scores in Economics by Causal Method from Brodeur et al. (2020)



(b) Distribution of Z scores for Publishability in Chopra et al. (2023)

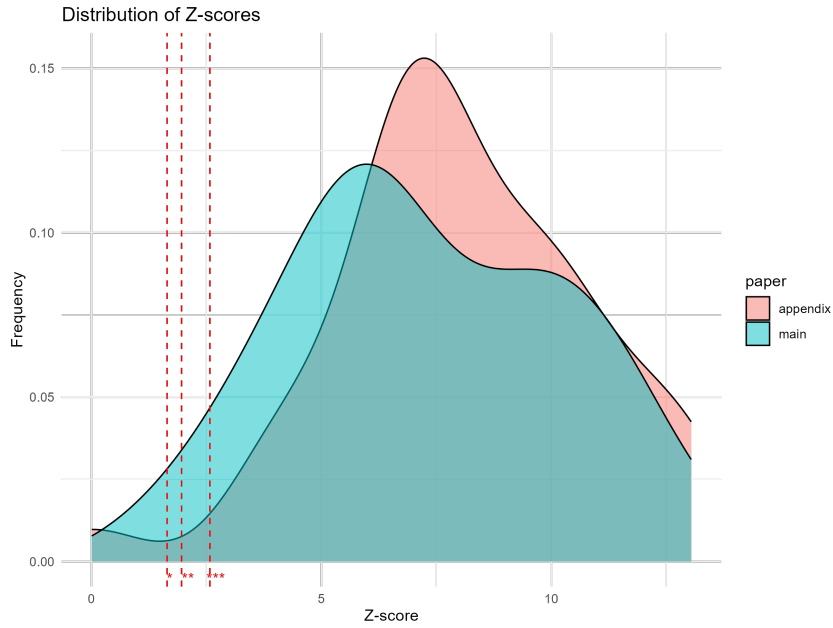
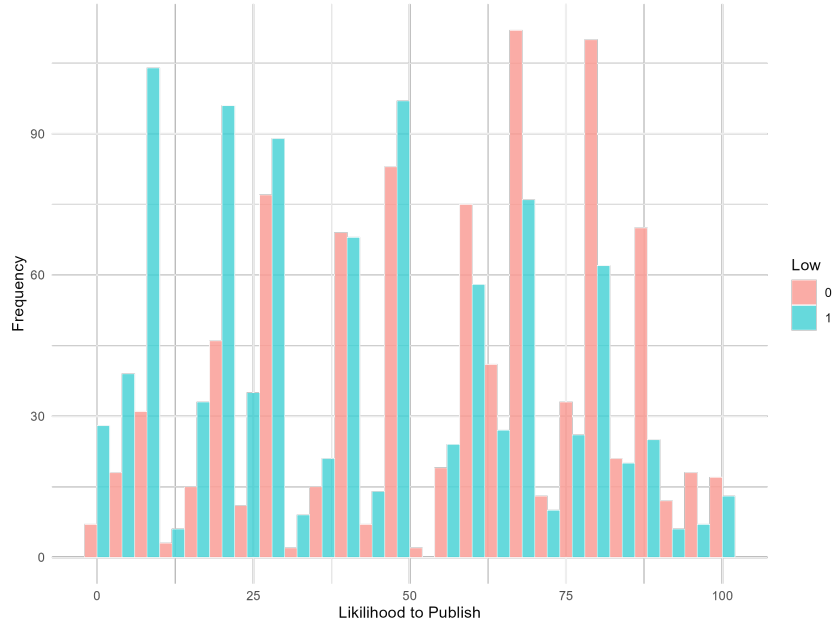


Figure 2: Comparison of Z scores in Chopra et al. (2023) to Brodeur et al. (2020)

Notes: This figure presents the distribution of z-scores presented in Chopra et al. (2023) using the point estimate and the standard errors presented. Main estimates are those that appear in the main text. Appendix estimates are those that appear in the supplementary online appendix. These z-scores are only for the primary outcome of interest: publishability of the presented vignette study. The vertical dashed lines represent the z-scores for statistical significance at * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$

(a) Distribution of Publishability by Treatment Status



(b) Reciprocal Distribution of Control Treatment on Publishability

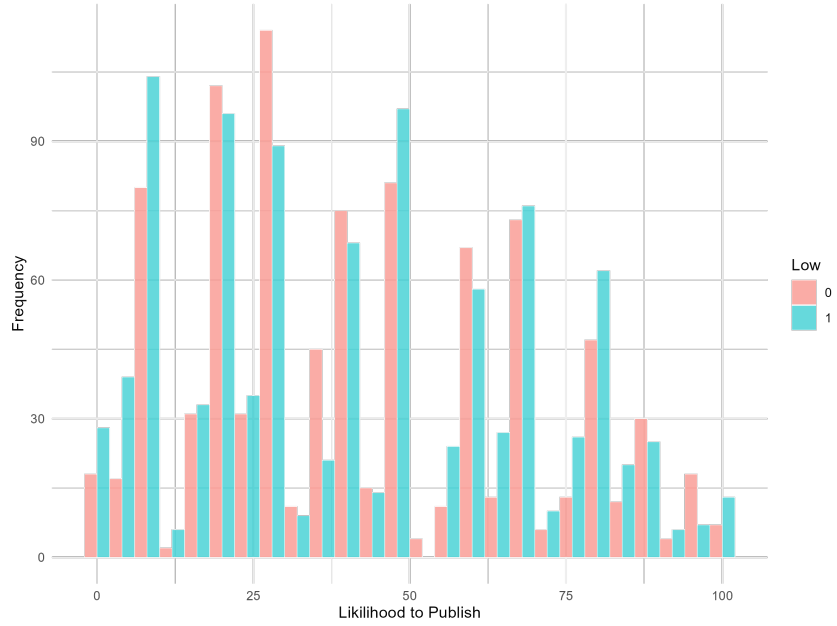


Figure 3: Comparison of Publishability Rating by Null Treatment

Notes: This figure presents two histograms of the primary outcome measure – publishability rating of the vignette studies – delimited by the null treatment effect status. The red bars indicate the control group receiving a statistically significant result and the blue bars indicate the treatment group receiving a statistically insignificant result. Sub-figure A presents that distribution as found in the data. Sub-figure B presents the distribution using the reciprocal distribution for the control group (e.g., $publish = 1 - publish$).